DEPARTMENT GUIDELINE

Guideline Title: **Potassium Imbalance**	Guideline Number: **GUID-3203-M006**

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature:	Signature:
Department Numbers: 3203	

I. **PURPOSE:** To recognize the patient with a potassium imbalance and to provide and maintain adequate oxygenation/ventilation, tissue perfusion, and cardiac rhythm. To provide emergent treatment, correct the imbalance and provide expeditious transport to the nearest appropriate medical facility.

II. **DEFINITIONS:**

a. Hypokalemia - a potassium (K+) serum level < 3.5 mEq/L
   i. Mild: 3.1-3.4 mEq/L, Moderate: 2.5-3 mEq/L, Severe: < 2.5 mEq/L

b. Hyperkalemia - a potassium (K+) serum level > 5.5 mEq/L
   i. Mild: 5.5 - 6 mEq/L, Moderate: 6.1 - 7 mEq/L, Severe: >7 mEq/L

III. **DEPARTMENT GUIDELINE:**

a. Establish care as defined by *GUID3203-G002 – General Patient Care.*

b. <u>Hyperkalemia:</u>
   i. Assess for clinical signs or symptoms of a hyperkalemic emergency
      1. Most serious: muscle weakness, paralysis, cardiac conduction abnormalities, cardiac arrythmias.
      2. Review ECG for: A tall peaked T wave with a shortened QT interval, followed by progressive lengthening of the PR interval and QRS duration. The P wave may disappear, and the QRS widens further to a sine wave. Conduction abnormalities, such as bundle branch blocks and arrhythmias (sinus bradycardia, sinus arrest, slow idioventricular rhythms, ventricular tachycardia, ventricular fibrillation, pseudo-ST-elevation myocardial infarction, pseudo-Brugada patterns, and asystole).
      3. Patients with moderate hyperkalemia (> 5.5 mEq/L) who have significant renal impairment and marked, ongoing tissue breakdown (e.g., rhabdomyolysis or crush injury, tumor lysis syndrome), ongoing potassium absorption (e.g., from substantial gastrointestinal bleeding), or a significant non-anion gap metabolic acidosis or respiratory acidosis.
      4. Consult with sending provider if patient has severe hyperkalemia with suspicious EKG changes. This needs to be addressed prior to departure for safe transport.
   ii. Calcium is indicated if K+ > 7 mEq/L or when ECG shows changes (loss of P waves & prolonged QRS duration.
      1. [PREFERED] **Calcium Gluconate 1-2 grams IV** over 3 min IV slowly from sending facility.

a. (If the patient is on digitalis, more caution is required as calcium can increase the toxic effects of digitalis. Add the appropriate dose to 100ml NS and infuse over 20-30 minutes.) **– OR –**
b. **Calcium Chloride 1 gram IV** slowly over 10 minutes, preferentially administered through central line or in the setting of acute decompensation/ cardiac arrest. Monitor for local irritation at the injection site.
1. If at sending facility, consider **Dextrose 50% 25 grams IV** followed by **Regular Insulin 5 units IV** to shift potassium into the intracellular space.
2. Check glucose level every 30 minutes and treat hypoglycemia per *GUID3203-M021 Hypoglycemia*.

iii. Consider **Sodium bicarbonate 8.4% 50 mEq IV/IO**.
iv. Consider **Albuterol 2.5-5 mg** in 3 ml of NS via nebulizer, may repeat.
v. If patient has signs of Diabetic Ketoacidosis, initiate therapy as outlined in *GUID3203-M002 – Adult Diabetic Ketoacidosis*

c. Hypokalemia:
i. If patient is severely bradycardic or manifesting cardiac arrhythmias, consider pharmacologic therapy or cardiac pacing per *GUID3203-C004 - Bradycardia and GUID3203-PR022 – Transcutaneous External Pacing*.
ii. If K+ level is < 2.5 mEq/L, consult sending/receiving physician for replacement orders. Consider **Potassium 10-20 mEq/100ml IV** at a rate not to exceed 10 mEq/hour through a peripheral IV access and 20mEq/h through a central venous access or IO.
iii. Serum K+ is difficult to replenish if serum magnesium is also low. Review Magnesium level and treat per *GUID3203-M005 Magnesium Imbalance*.

IV. **DOCUMENTATION:** EMS Charts

V. **REFERENCES:**

a. Littmann L, Gibbs MA. (2018). Electrocardiographic manifestations of severe hyperkalemia. *J Electrocardiol*, 51, 814.
b. Mount DB. Disorders of potassium balance. (2015). *In: Brenner and Rector's The Kidney*, *10th ed*, W.B. Saunders & Company, Philadelphia, p. 559.